

Proximal Gradient in the Dual

$$\min: f(x) + g(Ax)$$

$$\max: -f^*(-A^T z) - g^*(z) \quad \rightsquigarrow \min: \underbrace{f^*(-A^T z)} + \underbrace{g^*(z)}$$

$$z_{t+1} = \text{prox}_{\lambda g^*}(z_t - \eta \nabla(f^*(-A^T z)))$$

$$= \text{prox}_{\lambda g^*}(z_t + \eta A \underbrace{\nabla f^*(-A^T z)}_{\hat{x}})$$

$$= \text{prox}_{\lambda g^*}(z_t + \eta A \hat{x})$$

$$\text{where: } \hat{x} = \nabla f^*(-A^T z) \quad ; \quad \hat{x} \in \arg\min: f(x) + z^T A x.$$

$$\hat{x} = \arg\min: f(x) + z_t^T A x$$

$$z_{t+1} = \text{prox}_{\lambda g^*}(z_t + \eta A \hat{x})$$

Using the Moreau Decomposition

Recall:
$$x = \text{prox}_{\lambda h}(x) + \lambda \text{prox}_{\frac{1}{\lambda} h^*}(x/\lambda)$$

$$\begin{aligned} z_{t+1} &= \text{prox}_{\frac{1}{\lambda} g}(z_t + \lambda A \hat{z}) \\ &= z_t + \lambda A \hat{z} - \lambda \text{prox}_{\frac{1}{\lambda} g}(z_t/\lambda + A \hat{z}) \end{aligned}$$

$$= z_t + \lambda (A \hat{z} - \hat{y})$$

where:
$$\hat{z} = \arg\min: f(z) + z_t^T A z$$

Step 1 - Lag.

$$\hat{y} = \text{prox}_{\frac{1}{\lambda} g}(z_t/\lambda + A \hat{z})$$

$$\hat{y} = \arg\min_y: g(y) + \frac{\lambda}{2} \|A \hat{z} + z_t/\lambda - y\|_2^2$$

$$= \arg\min_y: g(y) + \underbrace{z_t^T (A \hat{z} - y)} + \frac{\lambda}{2} \|A \hat{z} - y\|_2^2$$

Step 2 - Arg Lag.

$$z_{t+1} = z_t + \lambda (A \hat{z} - \hat{y})$$

Step 3

Prox Grad vs PPA

$$\max: \underbrace{-f^*(-A^T z) - g^*(z)}$$

$$\text{PPA: } z_{t+1} = \text{prox}_{\lambda(\text{min})}^{(z_t)}$$

$$(\hat{x}, \hat{y}) = \underset{x, y}{\text{argmin}}: f(x) + g(y) + z^T(Ax - y) + \frac{\lambda}{2} \|Ax - y\|_2^2$$

$$z_{t+1} = z_t + \lambda (A\hat{x} - \hat{y}). \quad \text{No assumptions on } f$$

$$\text{Prox Grad: } \hat{x} = \underset{x}{\text{argmin}}: f(x) + z^T Ax$$

$$\hat{y} = \underset{y}{\text{argmin}}: g(y) + z^T (A\hat{x} - y) + \frac{\lambda}{2} \|A\hat{x} - y\|_2^2 \quad \text{f is strongly convex}$$

$$z_{t+1} = z_t + \lambda (A\hat{x} - \hat{y})$$

Accelerated Methods (FISTA)

Implemented exactly as above,
since we need a momentum form —
easy since we have the trajectory —
all still only involves gradients of f^*
which correspond to (partial) Lagrangian minimization.
↑
only over x .

Separable Structure, Comparison with ALM

$$\min: f_1(x_1) + f_2(x_2) + g(A_1 x_1 + A_2 x_2)$$

$$\max: -f_1^*(-A_1^T z) - f_2^*(-A_2^T z) - g^*(z)$$

Prox-Gradient-Ascent:

$$\hat{x}_1 = \arg\min: f_1(x_1) + z^T A_1 x_1$$

$$\hat{x}_2 = \arg\min: f_2(x_2) + z^T A_2 x_2$$

$$\begin{aligned} z_{t+1} &= \text{prox}_{\lambda g^*}(z_t + \lambda(A_1 \hat{x}_1 + A_2 \hat{x}_2)) \\ &= z_t + \lambda(A_1 \hat{x}_1 + A_2 \hat{x}_2 - \hat{y}) \end{aligned}$$

$$\hat{y} = \text{prox}_{\frac{1}{\lambda} g}(z_t / \lambda + A_1 \hat{x}_1 + A_2 \hat{x}_2)$$

$$= \arg\min_y: g(y) + \frac{\lambda}{2} \|y - (A_1 \hat{x}_1 + A_2 \hat{x}_2 + z_t / \lambda)\|^2$$

Requires: f_1^*, f_2^*
are smooth
i.e., f_1, f_2
strongly convex

Only Some Functions Smooth/Strongly Convex

$$\max: \underbrace{-f_1^*(-A_1^T z)}_{\text{smooth}} \underbrace{-f_2^*(-A_2^T z) - g^*(z)}_{\text{use prop of this}}$$

Exercise: $h(z) \triangleq f_2^*(-A_2^T z) + g^*(z)$

$$\Rightarrow \text{prox}_{\lambda h}(z) = z + \lambda(A_2 \hat{x}_2 - \hat{y})$$

$$(\hat{x}_2, \hat{y}) = \underset{(x_2, y)}{\text{argmin}}: f_2(x_2) + g(y) + z_c^T (A_2 x_2 - y) + \frac{\lambda}{2} \|A_2 x_2 - y\|_2^2$$

Dual Prox Grad: $\hat{x}_c = \underset{x_c}{\text{argmin}}: f_1(x_c) + z_c^T A_1 x_c$

$$(\hat{x}_2, \hat{y}) = \underset{(x_2, y)}{\text{argmin}}:$$

$$z_{t+1} = z_t + \lambda(A_1 \hat{x}_1 + A_2 \hat{x}_2 - \hat{y})$$